

## Education

**University of Massachusetts Amherst** (Manning CICS), *Ph.D. Computer Science* 2022 –  
*Supported by DOE Computational Science Graduate Fellowship, 2023 –*  
*Advised by: Mohammad Hajiesmaili and Prashant Shenoy*

**University of Massachusetts Amherst** 2019 – 2022  
*B.S. Computer Science & B.A. Political Science, summa cum laude*  
 Phi Beta Kappa, 21st Century Leader. GPA: 3.99 / 4.0  
*Honors Thesis (Advisor: Cameron Musco): Edge Dynamics and Opinion Polarization in Social Networks*

## Research Interests

I work at the intersection between theory and systems, encompassing the design of learning-augmented and provably robust algorithms for decision-making under uncertainty, and system designs that increase the flexibility, efficiency, and sustainability of computing infrastructure or energy systems.

## Experience

**Research Intern** **Redmond, Wash.**  
*Systems and Networking Group, Microsoft Research*  
 Summer 2026  
 ◦ Mentored by Behnaz Arzani, Pooria Namyar, and Konstantina Mellou

**Ph.D. Intern** **Golden, Colo.**  
*Power Systems Engineering Center, National Renewable Energy Laboratory (now NLR)*  
 Summer 2024  
 ◦ Advised by Joshua Comden and Andrey Bernstein

**Visiting Researcher** **Pasadena, Calif.**  
*Computing and Mathematical Sciences, California Institute of Technology*  
 Summer 2023  
 ◦ Advised by Adam Wierman

**Research Assistant** **Amherst, Mass.**  
*Manning College of Information and Computer Sciences*  
 Jun. 2022 – May 2023  
 ◦ Advised by Mohammad Hajiesmaili and Prashant Shenoy

## Awards and Honors

**Georgia Tech, UC Berkeley, U. of Michigan:** [2026 NextProf Nexus Workshop \(invited\)](#) Fall 2026  
**UMass Graduate School:** NSF Leadership Fellowship Fall 2025  
**Manning CICS:** Ph.D. Portfolio with Distinction Fall 2024  
**Manning CICS:** Dr. C. Mohan Graduate Scholarship Fall 2023  
**U. of Michigan, Georgia Tech, UC San Diego:** [2023 NextProf Pathfinder Workshop \(invited\)](#) Fall 2023  
**U.S. Department of Energy:** [Computational Science Graduate Fellowship](#) 2023-2027  
**Energy Transition Institute:** [NSF ELEVATE Fellowship](#) 2022-2024  
**UMass Amherst:** [Class of 2022 – 21st Century Leader Award \(highest undergraduate honor\)](#) Spring 2022  
**Manning CICS:** Outstanding Undergraduate Achievement Award Spring 2022  
**UMass Amherst:** [2021-2022 UMass Amherst Rising Researcher](#) Spring 2022  
**Manning CICS:** Outstanding Course Assistant Award Spring 2022  
**UMass Amherst:** Phi Beta Kappa Spring 2021

## Publications <sup>†</sup> – indicates undergraduate I advised (α–β) – indicates alphabetical author order

### Conference Publications

Mahsa Sahebdel, Ali Zeynali, Adam Lechowicz, Bo Sun, and Mohammad Hajiesmaili  
**Online Joint Pricing and Scheduling in Distributed Networked Platforms**  
*ACM Intl. Symp. on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks & Computing (MobiHoc), 2026.*

Adam Lechowicz, Rohan Shenoy<sup>†</sup>, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Christina Delimitrou  
**♠ SPADE: Signal-Aware DAG Scheduling and Dynamic Provisioning for Data Processing Clusters**  
*USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2026.

Adam Lechowicz, Nicolas Christianson, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy  
**Competitive and Learned Online Algorithms for HVAC Demand Management in Thermal Networks**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2026.

Ezra Johnson<sup>†</sup>, Adam Lechowicz, and Mohammad Hajiesmaili  
**Signal-Aware Workload Shifting Algorithms with Uncertainty-Quantified Predictors**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2026.

Cooper Sigrist, Anupama Sitaraman, Alice Zhang, Adam Lechowicz, Yuvraj Agarwal, and Mohammad Hajiesmaili  
**The Price to Site: Cost-Effective Incentives for Solar Deployment**  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2026.

Fabian Mkocheke, Walid A. Hanafy, Adam Lechowicz, David Irwin, and Prashant Shenoy  
**City-Scale Decarbonization of Residential Heating Under Transformer Constraints**  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2026.

Adam Lechowicz, Nicolas Christianson, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy  
**Online Smoothed Demand Management**  
*ACM SIGMETRICS*, 2026.

Cooper Sigrist, Archimedes Li, Pichsinee Lertsaroj, Ryan Boldi, Adam Lechowicz, Noman Bashir, and Mohammad Hajiesmaili  
**Multi-objective Evolutionary Learning for Near Pareto-Optimal Optimization of Solar Deployment (best paper candidate)**  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2025.

Adam Lechowicz, Rohan Shenoy<sup>†</sup>, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Christina Delimitrou  
**Carbon- and Precedence-Aware Scheduling for Data Processing Clusters (SIGCOMM short)**  
*ACM SIGCOMM*, 2025.

Mohammadreza Daneshvaramoli, Helia Karisani, Adam Lechowicz, Bo Sun, Cameron Musco, and Mohammad Hajiesmaili  
**Near-Optimal Consistency-Robustness Trade-Offs for Learning-Augmented Online Knapsack Problems**  
*International Conference on Machine Learning (ICML)*, 2025.

Adam Lechowicz, Joshua Comden, and Andrey Bernstein  
**Optimizing Individualized Incentives from Grid Measurements under Limited Knowledge of Agent Behavior**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2025.

Cooper Sigrist, Adam Lechowicz, Jovan Champ, Noman Bashir, and Mohammad Hajiesmaili  
**Lost in Siting: The Hidden Carbon Cost of Inequitable Residential Solar Installations**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2025.

Julia Köhlke, Adam Lechowicz, Oluwole Fabikun<sup>†</sup>, Noman Bashir, Abel Souza, Prashant Shenoy, and Sebastian Lehnhoff  
**Examining the Adoption of Electromobility Concepts Across Social Contexts for Energy Transition**  
*ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2024.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy  
**Chasing Convex Functions with Long-term Constraints**  
*International Conference on Machine Learning (ICML)*, 2024.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy  
**Online Conversion with Switching Costs: Robust and Learning-augmented Algorithms**  
*ACM SIGMETRICS / IFIP Performance*, 2024.

Roosbeh Bostandoost, Adam Lechowicz, Walid A. Hanafy, Noman Bashir, Prashant Shenoy, and Mohammad Hajiesmaili  
**LACS: Learning-Augmented Carbon-Aware Resource Scaling for Uncertain Demand**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2024.

Adam Lechowicz, Rik Sengupta, Bo Sun, Shahin Kamali, and Mohammad Hajiesmaili  
**Time Fairness in Online Knapsack Problems**  
*International Conference on Learning Representations (ICLR)*, 2024.

Adam Lechowicz, Noman Bashir, John Wamburu, Mohammad Hajiesmaili, and Prashant Shenoy  
**Equitable Network-Aware Decarbonization of Residential Heating at City Scale**  
*ACM International Conference on Future Energy Systems (e-Energy)*, 2023.

## Journal Publications

Nikita Bhalla, Adam Lechowicz, and Cameron Musco ( $\alpha-\beta$ )

### Local Edge Dynamics and Opinion Polarization

ACM Transactions on Intelligent Systems and Technology, August 2025. Also in WSDM '23.

Anupama Sitaraman, Adam Lechowicz, Noman Bashir, Xutong Liu, Prashant Shenoy, and Mohammad Hajiesmaili

### Dynamic Incentive Allocation for City-scale Deep Decarbonization

ACM Journal on Computing and Sustainable Societies, June 2025.

Adam Lechowicz, Nicolas Christianson, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy

### Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints

Proc. of the ACM on Measurement and Analysis of Computing Systems 9(1), Mar. 2025. Also in SIGMETRICS '25.

Adam Lechowicz, Nicolas Christianson, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, and Prashant Shenoy

### The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting

Proc. of the ACM on Measurement and Analysis of Computing Systems 7(3), Dec. 2023. Also in SIGMETRICS / Performance '24.

Bhawana Chhagiani, Camellia Zakaria, Adam Lechowicz, Jeremy Gummesson, and Prashant Shenoy

### FlowSense: Monitoring Airflow in Building Ventilation Systems Using Audio Sensing

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), March 2022.

## Refereed Workshop Papers

Roohbeh Bostandoost, Adam Lechowicz, Walid A. Hanafy, Prashant Shenoy, and Mohammad Hajiesmaili

### Quantifying the Carbon Reduction of DAG Workloads: A Job Shop Scheduling Perspective

International Workshop on Systems and Methods for Sustainable Large-Scale AI (GreenSys), 2026.

Cooper Sigrist, Archimedes Li, Pichsinee Lertsaroj, Ryan Boldi, Adam Lechowicz, Noman Bashir, and Mohammad Hajiesmaili

### Learning to Site: A Multi-objective Optimization of Rooftop Solar Installations using Evolutionary Neural Networks

ACM SIGMETRICS Performance Evaluation Review, August 2025.

Roohbeh Bostandoost, Walid A. Hanafy, Adam Lechowicz, Noman Bashir, Prashant Shenoy, and Mohammad Hajiesmaili

### Data-driven Algorithm Selection for Carbon-Aware Scheduling

ACM SIGEnergy Energy Informatics Review 4(5), Dec. 2024. Also in HotCarbon '24.

## Presentations

\* – indicates invited

### SPADE: Signal-Aware DAG Scheduling and Dynamic Provisioning for Data Processing Clusters

USENIX OSDI 2026

Jul. 2026

### Online Smoothed Demand Management

ACM SIGMETRICS 2026

Jun. 2026

### Learning-Augmented Algorithms for Spatiotemporal Online Allocation with Deadline Constraints

\* CS/OR/Econ Theory Lunch @ Yale ([Slides](#))

Mar. 2026

\* INFORMS Annual Meeting 2025 ([Slides](#))

Oct. 2025

ACM SIGMETRICS 2025

Jun. 2025

### Carbon- and Precedence-Aware Scheduling for Data Processing Clusters

ACM SIGCOMM 2025

Sep. 2025

### Learning-Augmented Online Optimization with Switching Costs and Long-term Constraints

\* INFORMS Applied Probability Society Conference 2025 ([Slides](#))

Jul. 2025

### Optimizing Individualized Incentives from Grid Measurements under Limited Knowledge of Agent Behavior

ACM e-Energy 2025

Jun. 2025

\* RSRG / FALCON Group Meeting @ Caltech ([Slides](#))

May 2025

ELEVATE Retreat, Energy Transition Institute @ UMass

Jan. 2025

\* ACM SIGEnergy Graduate Seminar ([Slides](#))

Nov. 2024

### Online Optimization with Switching Costs and Long-term Constraints

\* SODALab Group Meeting @ University of Alberta ([Slides](#))

Aug. 2024

### Online Conversion with Switching Costs: Robust and Learning-augmented Algorithms

ACM SIGMETRICS / IFIP Performance 2024

Jun. 2024

### The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting

ACM SIGMETRICS / IFIP Performance 2024

Jun. 2024

### Equitable Network-Aware Decarbonization of Residential Heating at City Scale

\* PIT x UMass: Developing Tech for the Public Interest ([poster in library exhibit](#))

Apr – May. 2024

|                                                                                                                                                                |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <i>ACM e-Energy 2023</i>                                                                                                                                       | Jun. 2023 |
| <b>Theoretical Foundations for Carbon-aware Load Shifting</b><br><i>ELEVATE Retreat, Energy Transition Institute @ UMass</i>                                   | Feb. 2024 |
| <b>Fresh Challenges for Online Problems: new directions motivated by practice</b><br>★ <i>RSRG / FALCON Lunch Seminar @ Caltech</i> ( <a href="#">Slides</a> ) | Jun. 2023 |
| <b>Local Edge Dynamics and Opinion Polarization</b><br><i>ACM WSDM 2023</i>                                                                                    | Feb. 2023 |
| <b>On the Necessary “Unfairness” of Competitive Online Algorithms</b><br><i>UMass CS Theory Seminar</i> ( <a href="#">Slides</a> )                             | Nov. 2022 |
| <b>Edge Dynamics and Opinion Polarization in Social Networks</b><br><i>Honors Thesis Defense</i> ( <a href="#">Slides</a> )                                    | Apr. 2022 |
| <b>Machine Learning for Absolute Beginners – Mathematical Foundations of ML</b><br><i>Workshop @ HackUMass IX</i> ( <a href="#">Slides</a> )                   | Nov. 2021 |

## Research Advising and Mentorship

|                                                                                                                 |             |
|-----------------------------------------------------------------------------------------------------------------|-------------|
| <b>Tri Than</b> ( <i>MS, UMass</i> )                                                                            | 2025 –      |
| <b>Urvi Gupta, Pihul Nyati, Samuel Nkemnkia, Rakshita Saroha</b> – <a href="#">ERSP @ CICS</a>                  | 2025 – 2026 |
| <b>Ezra Johnson</b> ( <i>UG, Caltech</i> ) – Computing for an Equitable Energy Transition REU                   | Summer 2025 |
| <b>Rohan Shenoy</b> ( <i>UG, UC Berkeley</i> ) – ( <i>received NSF GRFP</i> )                                   | 2024 – 2026 |
| <b>Anisha Prathi</b> ( <i>UG, UMass</i> ) – Undergraduate Honors Thesis                                         | 2023 – 2024 |
| <b>Kaosisochukwu Nwosu</b> ( <i>UG, UMass</i> ) – ( <i>received Outstanding Achievement Award</i> )             | 2023 – 2024 |
| <b>Lily Davoren</b> ( <i>UG, Bryn Mawr</i> ) – Computing for an Equitable Energy Transition REU                 | Summer 2023 |
| <b>Coo Katsuno, Saif Masoud, Riley Kim Connell</b> – <a href="#">URV Program @ CICS</a>                         | Summer 2023 |
| <b>Oluwole Fabikun</b> ( <i>UG, UMass</i> ) – LSAMP Scholar – ( <i>featured in graduating “Senior Series”</i> ) | 2023 – 2024 |

## Teaching

|                                                                                                                                                                                                                                                          |                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Guest Lecturer</b><br><i>INFO 203: A Networked World – College of Information and Computer Sciences</i><br>○ Presented two lectures on the impact of data centers and related algorithmic challenges in an undergraduate Informatics course at UMass. | <b>Amherst, Mass.</b><br>Spring 2026              |
| <b>Lecturer &amp; Mentor</b><br><a href="#">UMass Turing Summer Program</a> – College of Information and Computer Sciences                                                                                                                               | <b>Amherst, Mass.</b><br>Summers 2022, 2023, 2025 |
| <b>Course Assistant</b><br><i>COMPSCI 389: Intro to Machine Learning – College of Information and Computer Sciences</i>                                                                                                                                  | <b>Amherst, Mass.</b><br>Spring 2022              |

## Service

|                                                                                                                                                                                                                                                                                                                                                                                                               |                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>– Program Committees –</b><br><i>Shadow PC – ACM SIGMETRICS 2026</i>                                                                                                                                                                                                                                                                                                                                       | 2026                                                    |
| <b>– Workshop Organization –</b><br><a href="#">Workshop on Learning-augmented Algorithms: Theory and Applications (LATA)</a><br><a href="#">Workshop on Measurements, Modeling, &amp; Metrics for Carbon-Aware Computing (CarbonMetrics)</a><br><i>Chair – Workshop on Learning-augmented Algorithms: Theory and Applications (LATA)</i>                                                                     | SIGMETRICS '26<br>SIGMETRICS '25, '26<br>SIGMETRICS '25 |
| <b>– Journal Reviewing –</b><br><i>IEEE Transactions on Parallel and Distributed Systems</i><br><i>IEEE Transactions on Industrial Informatics</i><br><i>Sustainable Computing: Informatics and Systems</i><br><i>Computers and Operations Research</i><br><i>ACM Transactions on Modeling and Performance Evaluation of Computing Systems</i><br><i>IEEE Transactions on Network Science and Engineering</i> | 2026<br>2026<br>2026<br>2026<br>2026<br>2025            |

### – Conference Reviewing –

|                                                               |            |
|---------------------------------------------------------------|------------|
| European Symposium on Algorithms (ESA) (subreviewer)          | 2026       |
| International Conference on Learning Representations (ICLR)   | 2025, 2026 |
| Conference on Neural Information Processing Systems (NeurIPS) | 2025, 2026 |
| ACM SIGMETRICS (subreviewer)                                  | 2024, 2025 |

### – Department & University Service –

|                                                                                             |                       |
|---------------------------------------------------------------------------------------------|-----------------------|
| New Building Committee – Manning CICS                                                       | Jun. 2022 – ongoing   |
| Ph.D. Grad Representative – Manning CICS                                                    | Sep. 2025 – Sep. 2026 |
| Headmaster (Organizer) – <a href="#">OlympCICS</a> – Manning CICS                           | Jan. 2024             |
| ◦ Organized 2nd annual college-wide competition for CS PhD students.                        |                       |
| Grad Student Social Committee – Manning CICS                                                | Sep. 2023 – Aug. 2024 |
| New PhD Student Committee – Manning CICS                                                    | Sep. 2023 – Aug. 2024 |
| Student Trustee – Board of Trustees, University of Massachusetts                            | Jul. 2022 – Jul. 2023 |
| ◦ Student representative of the Amherst campus on system-wide Board of Trustees.            |                       |
| Committee Against Racism and for Equity – Manning CICS                                      | Sep. 2021 – Aug. 2022 |
| <a href="#">Flexible Learning Task Force &amp; Implementation Committee</a> – UMass Amherst | Jan. 2021 – May 2022  |

### – Outreach & Volunteering –

|                                                                                             |                       |
|---------------------------------------------------------------------------------------------|-----------------------|
| Judge – HackUMass XI & XIII (hackathon)                                                     | Nov. 2023, 2025       |
| Workshop Co-lead and Presenter – <a href="#">Holyoke Energy Justice Leaders Workshop #3</a> | May 2024              |
| Founder & Editor-in-Chief – UMass Index Yearbook                                            | Aug. 2020 – Jul. 2022 |
| ◦ Revived printed yearbook: proposed concept, raised funds, and delivered ~200 books.       |                       |
| Senator, Secretary of Technology – <a href="#">UMass Student Government Association</a>     | Sep. 2019 – May 2022  |

### – Seminars –

|                                                                                                  |                         |
|--------------------------------------------------------------------------------------------------|-------------------------|
| Grad Student Organizer – <a href="#">CS Systems &amp; Sustainability Lunches</a> , UMass Amherst | Fall 2024 – Spring 2025 |
| Grad Student Organizer – <a href="#">CS Theory Seminar</a> , UMass Amherst                       | Spring 2024 –           |
| Grad Student Organizer – Algorithms with Predictions Seminar, UMass Amherst                      | Fall 2022               |

## Open Source Projects

|                                                                                                                                                                                     |                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <a href="#">citation-checker</a> – command-line application, Python                                                                                                                 | Spring 2026            |
| Deterministic bibliography verifier for .bib files and PDFs that attempts to verify entries against databases such as DOI and CrossRef.                                             |                        |
| <a href="#">GlassTerm</a> – macOS application, Swift                                                                                                                                | Spring 2026            |
| My personal terminal app for macOS. Fast and lightweight with basic customization and tabs.                                                                                         |                        |
| <a href="#">Venti</a> – macOS application, Swift & JavaScript                                                                                                                       | Spring 2023, 2026      |
| Carbon-aware battery management tool for Apple silicon MacBooks. Prolongs battery health and defers charging to periods of time when grid electricity is sufficiently clean.        |                        |
| <a href="#">Backtrack</a> – iOS application, Swift & SwiftUI                                                                                                                        | Summer 2021            |
| Privacy-centric, open-source location logging solution that provides a history of location data for a personal device with minimal battery impact, leveraging deep API integration. |                        |
| <a href="#">Béton3 macro pad &amp; volume knob</a> – Hardware and firmware design, C, CAD, Arduino                                                                                  | Summer 2020            |
| Open-source playback control input device based on an Arduino Pro Micro.                                                                                                            |                        |
| <a href="#">UMass Campus Architecture</a> – iOS, watchOS application, SwiftUI                                                                                                       | Summer 2020, Fall 2025 |
| An interactive guide to highlight the architecture of the UMass campus.                                                                                                             |                        |

## Languages

English (U.S.) [native]; Polish [native]; American Sign Language [beginner]